

Cyclooxygenase inhibitory and antioxidant cyanidin glycosides in cherries and berries.

Anthocyanins from tart cherries, *Prunus cerasus* L. (Rosaceae) cv. Balaton and Montmorency; sweet cherries, *Prunus avium* L. (Rosaceae); bilberries, *Vaccinium myrtillus* L. (Ericaceae); blackberries, *Rubus* sp. (Rosaceae); blueberries var. Jersey, *Vaccinium corymbosum* L. (Ericaceae); cranberries var. Early Black, *Vaccinium macrocarpon* Ait. (Ericaceae); elderberries, *Sambucus canadensis* (Caprifoliaceae); raspberries, *Rubus idaeus* (Rosaceae); and strawberries var. Honeoye, *Fragaria x ananassa* Duch. (Rosaceae), were investigated for cyclooxygenase inhibitory and antioxidant activities. The presence and levels of cyanidin-3-glucosylrutinoside 1 and cyanidin-3-rutinoside 2 were determined in the fruits using HPLC. The antioxidant activity of anthocyanins from cherries was comparable to the commercial antioxidants, tert-butylhydroquinone, butylated hydroxytoluene and butylated hydroxyanisole, and superior to vitamin E, at a test concentration of 125 microg/ml. Anthocyanins from raspberries and sweet cherries demonstrated 45% and 47% cyclooxygenase-I and cyclooxygenase-II inhibitory activities, respectively, when assayed at 125 microg/ml. The cyclooxygenase inhibitory activities of anthocyanins from these fruits were comparable to those of ibuprofen and naproxen at 10 microM concentrations. Anthocyanins 1 and 2 are present in both cherries and raspberry. The yields of pure anthocyanins 1 and 2 in 100 g Balaton and Montmorency tart cherries, sweet cherries and raspberries were 21, 16.5; 11, 5; 4.95, 21; and 4.65, 13.5 mg, respectively. Fresh blackberries and strawberries contained only anthocyanin 2 in yields of 24 and 22.5 mg/100 g, respectively. Anthocyanins 1 and 2 were not found in bilberries, blueberries, cranberries or elderberries.